Galaxy Life and Death in One Step

Gas supply versus gas depletion timescales explain why galaxies stop and restart forming stars.

arXiv:2610.20505 — Step-function quenching from gas mass conservation, applied to ~350,000 galaxies.

Authors: Zhicheng He, Enci Wang, Luis C. Ho, Yingjie Peng, Huiyuan Wang, Yong Shi , Tao Wang, Yongquan Xue et al.

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Background

Galaxies quench when cold gas runs out, but the many proposed mechanisms—black holes, halos, morphology—have resisted a single unifying description.

> **No simple framework links diverse quenching processes to one observable, measurable quantity.**

What this paper is selling

Gas mass conservation yields a step-function quenching response set by supply versus depletion timescales. Fitting the quenched fraction then lets observers infer the gas-supply timescale directly.

Key results to remember

- Characteristic gas-supply timescale is about 1 Gyr across the low-redshift sample.
- Black holes drive inside-out quenching; dark matter halos drive outside-in quenching.
- Intermediate surface-density galaxies sit at a critical state for quenching and rejuvenation.

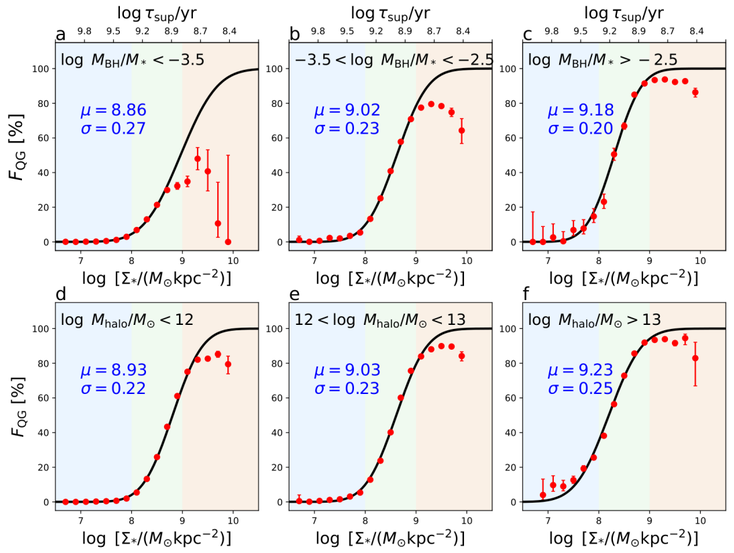


Figure 2 | Black hole growth lengthens and narrows the supply-timescale distribution, while halo mass lengthens and broadens it.

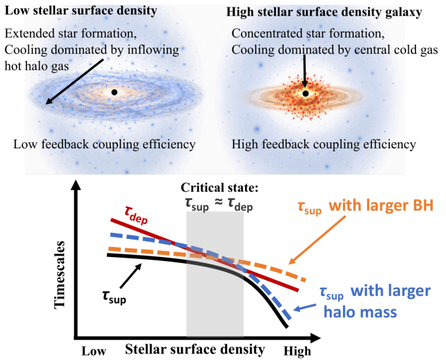


Figure 4 | A physical picture: halo cooling sets supply at low density, central concentration at high density, with critical balance between.